

VICTOR CHISOM MUOJEKE

Master Sprint Plan — 4 Projects — April 28 to December 19, 2026

Start fresh · Build with Claude Code · MASc alongside

Project 1 — MUNext v2 Enterprise

April 28 – June 30

Build MUNext from scratch. New repo. Next.js 16 + TypeScript frontend consuming a restructured modular Laravel 12 API. Add Pest testing. Containerise with Docker. This project is the foundation — everything else builds on top of it.

Week 1 — Environment + Laravel Scaffold

Apr 28 – May 4

Apr 28 Mon	Create new GitHub repo: munext-v2. Init Laravel 12 project. Configure .env, PostgreSQL connection, run migrations. Confirm app boots.	<i>Watch first: Laravel 12 install (10 min)</i>
Apr 29 Tue	Install Pest. Write your first 3 Pest tests for the health check and auth endpoints. Run the test suite and make them pass.	<i>Watch first: Pest for Laravel (15 min)</i>
Apr 30 Wed	Restructure Laravel into modular architecture: create Modules/Auth, Modules/Jobs, Modules/Students folders. Move existing models and controllers.	<i>Watch first: Modular Laravel (15 min)</i>
May 1 Thu	Build the Auth module: register, login, logout endpoints with Sanctum. Write Pest tests for all three. All tests green.	
May 2 Fri	Build the Jobs module: CRUD endpoints for job postings. Write Pest tests for create, read, update, delete.	
May 3 Sat	Build the Students module: profile CRUD, skills array, GPA, graduation year. Full Pest test coverage.	
May 4 Sun	API review day. Confirm all modules work together. Write a Postman collection. Document all endpoints in README.	

Week 2 — Next.js 16 + TypeScript Setup

May 5 – May 11

May 5 Mon	Init Next.js 16 project with TypeScript: <code>npx create-next-app@latest munext-frontend --typescript</code> . Configure Tailwind CSS. Confirm it boots.	<i>Watch first: Next.js in 15 min (watch before starting)</i>
May 6 Tue	Set up folder structure: <code>app/</code> , <code>components/</code> , <code>lib/</code> , <code>types/</code> . Create TypeScript interfaces for Job, Student, User. Configure environment variables.	<i>Watch first: TypeScript interfaces (10 min)</i>
May 7 Wed	Build API client layer in <code>lib/api.ts</code> using <code>fetch</code> + TypeScript generics. Type all Laravel API responses. Handle errors gracefully.	
May 8 Thu	Build Auth flow: login page, register page, session handling with NextAuth or Sanctum cookie. Protect routes with middleware.	<i>Watch first: NextAuth basics (15 min)</i>
May 9 Fri	Build the Jobs listing page: fetch from Laravel API, display with TypeScript types, loading states, error boundaries.	
May 10 Sat	Build the Student dashboard: profile view, skills display, GPA. Wire to Laravel API. All data TypeScript typed.	
May 11 Sun	School starts. Review week. Fix any rough edges. Commit clean state to GitHub. From here: 2hrs weekday, 3hrs weekend.	

Weeks 3–6 — Core Features + TanStack Query

May 12 – June 8

Week 3	Install TanStack Query. Migrate all API calls from raw <code>fetch</code> to <code>useQuery</code> hooks. Add caching, background refetch, and loading states across all pages.	<i>Watch first: TanStack Query intro (15 min)</i>
---------------	---	---

Week 4	Build Employer dashboard: post jobs, view applicants, see skill match scores. All data fetched via TanStack Query with proper TypeScript types.	
Week 5	Build the ML skill matching endpoint in Laravel (TF-IDF service already exists — integrate it into the Jobs module properly). Pest tests for the matcher.	
Week 6	Build the skill match results page in Next.js: show composite score, matched skills, missing skills per student. Recharts bar chart for skill overlap.	<i>Watch first: Recharts intro (10 min)</i>

Weeks 7–8 — Docker + Project 1 Complete	June 9 – June 30
--	------------------

Week 7	Write Dockerfile for Laravel API. Write Dockerfile for Next.js frontend. Write docker-compose.yml that runs both + PostgreSQL together. Test locally.	<i>Watch first: Docker in 20 min (watch before)</i>
Week 8	Final integration testing. All Pest tests passing. Docker compose up works cleanly. Write architecture README with diagram. Tag v1.0 release on GitHub.	

Project 2 — DataFlow Analytics Engine	July – August 8
--	-----------------

Build a PySpark data pipeline that ingests job and student data, transforms it via Spark SQL, and feeds aggregated analytics back to the Laravel API. Add a live Recharts dashboard to MUNext. Use Laravel Horizon to queue and schedule Spark jobs.

Week 1 — PySpark Setup + First Pipeline	July 1 – July 7
--	-----------------

Day 1–2	Install PySpark locally. Write a Python script that reads job postings CSV, loads into a Spark DataFrame, and runs basic Spark SQL queries (count by category, avg salary).	<i>Watch first: PySpark in 20 min (watch before Day 1)</i>
Day 3–4	Connect PySpark to PostgreSQL using JDBC driver. Read real MUNext jobs table into a Spark DataFrame. Run aggregation queries.	
Day 5–7	Build the first full pipeline: ingest jobs + students data → Spark SQL transforms → write aggregated output back to a PostgreSQL analytics table.	

Week 2 — Laravel Horizon + Job Queues	July 8 – July 14
--	------------------

Day 1–2	Install Laravel Horizon. Configure Redis. Create a SparkPipelineJob that triggers the Python PySpark script via shell exec. Queue it.	<i>Watch first: Laravel Horizon intro (15 min)</i>
Day 3–4	Add job scheduling: run SparkPipelineJob every hour via Laravel scheduler. Monitor jobs in the Horizon dashboard at /horizon.	
Day 5–7	Build analytics API endpoints in Laravel: GET /api/analytics/skills-demand, /api/analytics/hiring-trends, /api/analytics/salary-distribution.	

Week 3 — Recharts Dashboard in Next.js	July 15 – July 21
---	-------------------

Day 1–2	Build the Analytics Dashboard page in Next.js. Fetch from the 3 analytics endpoints. Display loading skeletons while data loads.	
Day 3–4	Add Recharts: BarChart for skills demand, LineChart for hiring trends over time, PieChart for job categories.	
Day 5–7	Polish: date range filters, export to CSV button, responsive layout. Add Pest tests for all 3 analytics endpoints.	

Weeks 4–5 — Integration + Project 2 Complete	July 22 – August 8
---	--------------------

Week 4 Full integration testing: Spark pipeline runs, Horizon queues it, analytics endpoints return fresh data, Next.js dashboard displays it. Fix all issues.

Week 5 Update Docker compose to include Redis + Horizon worker. Update README. Tag v2.0. Summer break starts Aug 9 — go hard.

Project 3 — AI-Powered Job Matcher

August 9 – September 7

Summer break — 5 hrs/day, 4 weeks. Upgrade the skill matcher with a LangChain RAG pipeline. CV text in, Spark-processed job data retrieved as context, LLM streams a match analysis. Add Vercel AI SDK for streaming UI. Full audit logging.

Week 1 — LangChain + RAG Foundation

Aug 9 – Aug 15

Day 1–2 Install LangChain Python. Understand the RAG pattern: Document → Embeddings → Vector Store → Retriever → LLM chain. Build a hello-world RAG that answers questions about a text file. *Watch first: LangChain RAG in 20 min (watch before Day 1)*

Day 3–4 Load MUNext job postings from PostgreSQL into LangChain Documents. Generate embeddings using OpenAI or Anthropic. Store in a vector store (ChromaDB locally).

Day 5–7 Build the core RAG chain: student CV text → retriever finds top matching jobs from vector store → LLM generates a match analysis. Test with 5 sample CVs.

Week 2 — Laravel API + LangChain Service

Aug 16 – Aug 22

Day 1–2 Wrap the LangChain pipeline in a FastAPI microservice: POST /match with CV text, returns streaming LLM response. Test with curl. *Watch first: FastAPI basics (10 min)*

Day 3–4 Add a Laravel endpoint that calls the FastAPI service: POST /api/ai/match. Proxy the streaming response. Add Sanctum auth. Write Pest tests.

Day 5–7 Add Spatie Activity Log to record every AI match request: who called it, what CV was submitted, what tokens were used, timestamp.

Week 3 — Vercel AI SDK + Streaming UI

Aug 23 – Aug 29

Day 1–2 Install Vercel AI SDK in Next.js. Understand useChat and useCompletion hooks. Build a simple streaming chat component first as a learning exercise. *Watch first: Vercel AI SDK intro (15 min)*

Day 3–4 Build the AI Match page: student uploads/pastes CV text, clicks Analyse, streaming response appears word-by-word from the LangChain pipeline.

Day 5–7 Add match history page: list past AI match results pulled from Spatie activity log. Show token usage, date, match score summary.

Week 4 — Polish + Project 3 Complete

Aug 30 – Sep 7

Day 1–3 Full integration test: upload CV → Laravel calls FastAPI → LangChain retrieves jobs → LLM streams → Next.js displays → Spatie logs. Fix all issues.

Day 4–5 Update Docker compose to include the FastAPI service. Update README with RAG architecture diagram. Tag v3.0.

Day 6–9 Buffer days. Fall semester starts Sep 8. Capstone begins. Use these days to wrap cleanly and prepare your head for the shift in pace.

Project 4 — Cloud Deploy + Portfolio Polish

September 8 – December 19

Capstone is running. Light pace — 1 hr weekday, 2–3 hrs weekend. Provision DigitalOcean via Terraform. Deploy on DOKS Kubernetes. Polish CV, LinkedIn, and build the public portfolio. December 19 — done.

Weeks 1–2 — Terraform + DigitalOcean Basics

Sep 8 – Sep 21

- | | | |
|---------------|--|---|
| Week 1 | Install Terraform. Read DigitalOcean provider docs. Write a main.tf that provisions one Droplet and one managed PostgreSQL database. Run terraform apply, confirm it works, run terraform destroy. | <i>Watch first: Terraform intro (20 min — watch before Day 1)</i> |
| Week 2 | Extend Terraform config: add a DigitalOcean Spaces bucket (file storage), a domain record, and a firewall. Apply, verify, destroy. Build the muscle memory of always destroying. | |

Weeks 3–5 — DOKS Kubernetes

Sep 22 – Oct 12

- | | | |
|---------------|--|--|
| Week 3 | Provision a DOKS cluster via Terraform (1 node, smallest size). Connect kubectl to it. Deploy a hello-world pod. Confirm it runs. Destroy. | <i>Watch first: Kubernetes basics (20 min)</i> |
| Week 4 | Write Kubernetes manifests: Deployment + Service for Laravel API. Deploy to DOKS. Expose via LoadBalancer. Hit the endpoint. Destroy. | |
| Week 5 | Write K8s manifests for Next.js frontend and FastAPI service. Deploy all three together. Confirm they talk to each other. Destroy. | |

Weeks 6–9 — Full Platform Deployment

Oct 13 – Nov 9

- | | | |
|---------------|--|---|
| Week 6 | Deploy managed PostgreSQL via Terraform. Connect Laravel API pod to it via K8s secrets. Run migrations inside the cluster. Confirm data flows. | |
| Week 7 | Add GitHub Actions CI/CD: on push to main, build Docker images, push to DigitalOcean Container Registry, deploy to DOKS automatically. | <i>Watch first: GitHub Actions (15 min)</i> |
| Week 8 | Deploy Redis + Laravel Horizon worker as K8s Deployment. Confirm Spark job queue still works end-to-end in the cluster. | |
| Week 9 | Full end-to-end test: create student, post job, run AI match, check analytics dashboard — all running on DOKS. Fix issues. | |

Weeks 10–15 — Portfolio Polish + Finish Line

Nov 10 – Dec 19

- | | | |
|----------------|--|--|
| Week 10 | Write the architecture README: system diagram in Mermaid, component explanations, tech stack breakdown, deployment instructions. | |
| Week 11 | Record Loom demo videos: 3 min walkthrough of each project (4 videos total). Link from GitHub README and LinkedIn. | |
| Week 12 | One targeted open-source PR: find a Laravel, Next.js, or LangChain repo with a good-first-issue. Submit a real, quality contribution. | |
| Week 13 | Update CV with all new skills. Update LinkedIn headline, about section, and all 4 projects with descriptions and live URLs. | |
| Week 14 | Final platform review. All services deployed and healthy. Tag v4.0. Write a personal retrospective: what you learned, what you'd do differently. | |
| Week 15 | Buffer + graduation prep. December 19 — roadmap complete. | |

SUMMARY — 4 PROJECTS AT A GLANCE

Project	Period	Key Deliverable	New Skills
1 — MUNext v2	Apr 28–Jun 30	Next.js + Laravel modular + Docker	Next.js, TypeScript, Pest, Modular Laravel
2 — DataFlow	Jul 1–Aug 8	PySpark pipeline + Recharts dashboard	Spark, PySpark, Horizon, Recharts
3 — AI Matcher	Aug 9–Sep 7	LangChain RAG + streaming UI	LangChain, Vercel AI SDK, RAG, FastAPI
4 — Cloud Deploy	Sep 8–Dec 19	DOKS + Terraform + CI/CD + portfolio	Terraform, Kubernetes, DigitalOcean, IaC

NON-NEGOTIABLES — READ THESE EVERY MONDAY MORNING

- Watch max 20 minutes before building. Then close the video and open VS Code.
- Claude Code assists — you stay involved. Understand every line before moving on.
- Capstone always wins in Window 4. Never let a project task delay your academic work.
- Destroy DigitalOcean resources at the end of every session. No exceptions.
- Update your CV and LinkedIn after each project completes — not all at December.
- If a week slips due to exams, don't try to catch up — just continue from where you are.

Victor Chisom Muojeke · Master Sprint Plan · 4 Projects · Dec 19, 2026